

Past Paper ← (2015) →

①

Q3

~~Q1~~

X	P(X)	X P(X)	X² P(X)
4	0.25	1	4
5	0.35	1.75	8.75
6	0.30	1.8	10.8
7	0.10	0.7	4.9
		5.25	28.45

$$\mu_{\bar{x}} = E(\bar{x}) = \sum \bar{x} P(\bar{x})$$

$$\mu_{\bar{x}} = \mu$$

$$\mu_{\bar{x}} = 5.25$$

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n}$$

$$\begin{aligned} \sigma_{\bar{x}}^2 &= \text{Var}(\bar{x}) = \sum \bar{x}^2 P(\bar{x}) - (\mu_{\bar{x}})^2 \\ &= 28.45 - (5.25)^2 \\ &= 0.8875 \end{aligned}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$P(\bar{x} < 5.5)$$

$$= P\left(\frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} < \frac{5.5 - \mu_{\bar{x}}}{\sigma_{\bar{x}}} \right)$$

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Past Paper (2015) →

Q3

<u>X</u>	<u>P(X)</u>	<u>X P(X)</u>	<u>X² P(X)</u>
4	0.25	1	4
5	0.35	1.75	8.75
6	0.30	1.8	10.8
7	0.10	0.7	4.9
		5.25	28.45

$$n = 25$$

$$\mu = E(X) = \sum X P(X)$$

$$\mu = 5.25$$

$$\begin{aligned} \sigma^2 &= \sum X^2 P(X) - (\mu)^2 \\ &= 28.45 - (5.25)^2 \\ &= 0.8875 \end{aligned}$$

$$\sqrt{\sigma^2} = \sqrt{0.8875}$$

$$\sigma = 0.9421$$

According to the Property
Sampling Distribution of Sample mean ($\bar{x}$)

$$\Rightarrow \mu_{\bar{x}} = \mu$$

$$\boxed{\mu_{\bar{x}} = 5.25}$$

$$\Rightarrow \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

$$\sigma_{\bar{x}} = \frac{0.9421}{\sqrt{25}}$$

(11)

(2018)

(3)

$$\sigma_{\bar{x}} = 0.1884$$

$$P(\bar{x} < 5.5)$$

$$P\left(\frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} < \frac{5.5 - \mu_{\bar{x}}}{\sigma_{\bar{x}}}\right)$$

$$P(Z < \frac{5.5 - 5.25}{0.1884})$$

$$P(Z < 1.32696)$$

$$P(\Phi(1.32696))$$

$$= 0.90658$$

(2018) -

(4)

Q5
100

$$n = 500$$

$$x = 43$$

$$1 - \alpha = 90\%$$

$$1 - \alpha = 0.9$$

$$\alpha = 0.1$$

$$\frac{\alpha}{2} = \frac{0.1}{2}$$

$$\frac{\alpha}{2} = 0.05$$

$$\frac{1 - \alpha}{2} = \frac{1 - 0.05}{2}$$

$$\frac{1 - \alpha}{2} = 0.95$$

$$Z_{\frac{1 - \alpha}{2}} = 0.950$$

$$Z_{\frac{1 - \alpha}{2}} = 1.6449$$

$$P = \frac{x}{n}$$

$$P = \frac{43}{500}$$

$$P = 0.086$$

$$q = 1 - P$$

$$q = 1 - 0.086$$

$$q = 0.914$$

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$$P \pm Z_{\text{Tab}} \sqrt{\frac{pq}{n}}$$

$$0.086 \pm (1.6449) \sqrt{\frac{(0.086)(0.914)}{500}}$$

$$0.086 \pm (1.6449)(0.0125)$$

$$0.086 \pm 0.0206$$

$$0.0654 < P < 0.1066$$

$$0.0654 < P < 0.1066$$

Q5
(b)

$$\bar{X} = 42, S = 10, n = 25$$

$$1 - \alpha = 98\%$$

$$1 - \alpha = 0.98$$

$$\alpha = 0.02$$

$$\frac{\alpha}{2} = \frac{0.02}{2}$$

$$\frac{\alpha}{2} = 0.01$$

$$1 - \frac{\alpha}{2} = 0.99$$

$$df = v = n - 1$$

$$= 25 - 1$$

$$v = 24$$

$$\chi^2_{\frac{\alpha}{2}, v} = 10.86$$

$$\chi^2_{1 - \frac{\alpha}{2}, v} = 42.98$$

— (2015) —

⑥

$$\begin{aligned}\sum (x - \bar{x})^2 &= n s^2 \\ &= (25)(10)^2 \\ &= 2500\end{aligned}$$

Now,

Two sided C-I for σ^2 is

$$\frac{\sum (x - \bar{x})^2}{\chi^2_{1-\frac{\alpha}{2}, n}} < \sigma^2 < \frac{\sum (x - \bar{x})^2}{\chi^2_{\frac{\alpha}{2}, n}}$$

$$\frac{2500}{42.98} < \sigma^2 < \frac{2500}{10.86}$$

$$58.1666 < \sigma^2 < 230.2025$$

Q6

$$n_1 = 12$$

$$n_2 = 20$$

$$\bar{x}_1 = 10$$

$$\bar{x}_2 = 18$$

$$s_1^2 = 1200$$

$$s_2^2 = 1000$$

$$\alpha = 5\%$$

Step (1)

$$H_0 : \mu_1 - \mu_2 \geq 10$$

$$H_1 : \mu_1 - \mu_2 < 10$$

(Lower tail Test)

Step (2)

Level of Significance

$$\alpha = 5\%$$

$$\alpha = 0.05$$

(2015)

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Step (3)

Test Statistic:-

$$T = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{Sp^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

Step (4)

Critical Region

$$\begin{aligned} df = v &= n_1 + n_2 - 2 \\ &= 12 + 20 - 2 \\ &= 30 \end{aligned}$$

$$T_{\alpha, v} = T_{0.05, 30} = -1.697$$

Step (5)

Decision Rule

We will reject H_0 if

$$T_{cal} < -1.697$$

Step (6)

Calculation

$$Sp^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(12 - 1)(1200) + (20 - 1)(1000)}{12 + 20 - 2}$$

$$Sp^2 = 1073.33$$

(2015)

Q. 9

$$T_{cal} = (10 - 18) - 10$$

$$\sqrt{(1073.33) \left(\frac{1}{12} + \frac{1}{20} \right)}$$

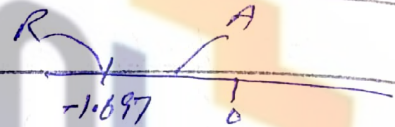
$$T_{cal} = -1.505$$

Step (1)

Conclusion

$$-1.505 < -1.697$$

We do not reject H_0 .



Q7

$$n_1 = 11$$

$$n_2 = 6$$

$$\bar{x}_1 = 50.3$$

$$\bar{x}_2 = 37.8$$

$$s_1^2 = 46.74$$

$$s_2^2 = 51.47$$

$$\alpha = 5\%$$

Step (1)

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

Step (2)

Level of Significance

$$\alpha = 5\%$$

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

→ (2015)

(10)

Step (3)

Test statistical,

$$F = \frac{s_2^2}{s_1^2}$$

Step (4)

Critical region,

$$df = v_1 = 11 - 1 = 10$$

$$v_2 = 6 - 1 = 5$$

$$F_{0.025, (5, 10)}$$

$$F_{0.025, (5, 10)} = 4.24$$

$$\frac{1}{F_{0.025, (10, 5)}} = \frac{1}{9.89} = 0.10$$

Step (5) Decision

We will reject H_0 if

$$F_{cal} > 4.24$$

$$F_{cal} < 0.10$$

Step (6)

$$F_{cal} = \frac{s_2^2}{s_1^2} = \frac{51.47}{46.74}$$

$$F_{cal} = 1.10$$

Step (7) Conclusion

$$1.10 > 4.24$$

$$1.10 < 0.10$$

we do not reject H_0

Past Paper (2017)

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Q4

$$N=9, \quad n=6$$

2 values drawn from each stratum (equal allocation)

Stratum I

Stratum II

Stratum III

1

6

11

2

8

15

4

16

19

2 values from each stratum.

$$n=6$$

n^m

$$N_1 = \frac{N}{\binom{C}{n}} = \frac{3}{\binom{C}{2}} = 3$$

$$N_2 = \frac{N}{\binom{C}{n}} = \frac{2}{\binom{C}{2}} = 1$$

$$N_3 = \frac{N}{\binom{C}{n}} = \frac{4}{\binom{C}{2}} = 6$$

Total number of samples

$$3 \times 1 \times 6 = 18$$

(2017)

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Stratum I
Samples

Stratum II
Samples

Stratum III
Samples

(1,2)

(6,8)

(11,15)

(1,4)

(11,16)

(2,4)

(11,19)

(15,16)

(15,19)

(16,19)

Stratified Samples of sample size = 6

Samples

$\bar{X}$

1,2,6,8,11,15

$$2x_{11} = 43/6 = 7.17$$

1,2,6,8,11,16

$$= 7.33$$

1,2,6,8,11,19

$$= 7.83$$

1,2,6,8,15,16

$$= 8$$

1,2,6,8,15,19

$$= 8.50$$

1,2,6,8,16,19

$$= 8.67$$

1,4,6,8,11,15

$$7.5$$

1,4,6,8,11,16

$$7.67$$

1,4,6,8,11,19

$$8.17$$

1,4,6,8,15,16

$$8.33$$

1,4,6,8,15,19

$$8.83$$

1,4,6,8,16,19

$$9$$

(2017)

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Samples

$\bar{x}$

2, 4, 6, 8, 11, 15

7.67

2, 4, 6, 8, 11, 16

7.83

2, 4, 6, 8, 11, 19

8.33

2, 4, 6, 8, 15, 16

8.50

2, 4, 6, 8, 15, 19

9

2, 4, 6, 8, 16, 19

9.17

Q5

<u>100/class</u>	<u>Number / f</u>	<u>x</u>	<u>fx</u>	<u>fx²</u>
80-89	30	84.5	2535	214207.5
90-99	52	94.5	4914	464373
100-109	75	104.5	7837.5	819018.75
110-119	109	114.5	12480.5	1429017.25
120-129	65	124.5	8092.5	1007816.25
130-139	42	134.5	5649	759790.5
140-149	27	144.5	3901.5	563766.75
	$\Sigma f = 400$		$\Sigma fx = 45410$	$\Sigma fx^2 = 5257690$

$\Sigma fx = 45410$
 $\Sigma fx^2 = 5257690$

$$\sum (x - \bar{x})^2$$

$$= \sum f x^2 - \frac{(\sum f x)^2}{\sum f}$$

$$= 5257690 - \frac{(45410)^2}{400}$$

$$=$$

$$S = \sqrt{\frac{\sum f (x - \bar{x})^2}{\sum f - 1}}$$

$$\text{Standard Error} = \frac{S}{\sqrt{n}} = \frac{S}{\sqrt{400}}$$

$$\frac{\sum f x}{\sum f} = \frac{45410}{400} = 113.525$$

$$\sum f (x - \bar{x})^2 = \sum f x^2 - \frac{(\sum f x)^2}{\sum f}$$

$$= 5257690 - \frac{(45410)^2}{400}$$

$$\sum f (x - \bar{x})^2 = 102519.75$$

$$S = \sqrt{\frac{\sum f (x - \bar{x})^2}{\sum f - 1}} = \sqrt{\frac{102519.75}{399}}$$

$$S = 16.03$$

(2017) -

(15)

$$\text{Standard Error} = \sigma_{\bar{x}} = \frac{S}{\sqrt{n}} \quad n = 24$$

$$\sigma_{\bar{x}} = \frac{16.03}{\sqrt{400}}$$

$$\sigma_{\bar{x}} = 0.8015$$

Q7

$$n_1 = 332$$

$$n_2 = 615$$

$$\bar{x}_1 = 12.78$$

$$\bar{x}_2 = 12.99$$

$$s_1 = 2.43$$

$$s_2 = 2.48$$

Step 1

$$H_0: \mu_2 - \mu_1 = 0$$

$$H_1: \mu_2 - \mu_1 \neq 0$$

Step 2

Level of Significance :-

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

$$1 - \frac{\alpha}{2} = 0.975$$

Step 3

$$Z = \frac{(\bar{x}_2 - \bar{x}_1) - (\mu_2 - \mu_1)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Step 4

$$Z = \frac{12.99 - 12.78}{\sqrt{\frac{(2.43)^2}{332} + \frac{(2.48)^2}{615}}}$$

$$\sqrt{\frac{(2.43)^2}{332} + \frac{(2.48)^2}{615}}$$

— (2017) —

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$$Z = 1.2598$$

Step (5)

Critical region

γ

$$Z_{\text{tab}} = \pm 1.9600$$

Step (6)

We will reject H_0 if

$$Z_{\text{cal}} > 1.96, \quad Z_{\text{cal}} < -1.96$$

Step (7)

Conclusion

PUACP

$$1.2598 > 1.96, \quad 1.2598 < -1.96$$

We do not reject H_0 .

γ

α_2

(2017)-

(17)

Q8

Before	After	$D_{x_2 - x_1}$	$\frac{D_{x_2 - x_1}^2}{}$
44	53	9	81
40	38	-2	4
61	69	8	64
52	57	5	25
32	46	14	196
44	39	-5	25
70	73	3	9
41	48	7	49
67	73	6	36
72	74	2	4
53	60	7	49
72	78	6	36
		$\Sigma D = 60$	$\Sigma D^2 = 578$

$$\bar{D} = \frac{\Sigma D}{n} = \frac{60}{12} = 5$$

$$\Sigma(D - \bar{D})^2 = \Sigma D^2 - \left(\frac{\Sigma D}{n}\right)^2$$

$$= 578 - \frac{(60)^2}{12}$$

$$\Sigma(D - \bar{D})^2 = 278$$

$$S_D = \sqrt{\frac{\Sigma(D - \bar{D})^2}{n-1}}$$

$$S_D = \sqrt{\frac{278}{11}}$$

$$S_D = 5.03$$

Now,

Step (1)

$$H_0: \mu_2 \leq \mu_1$$

$$H_1: \mu_2 > \mu_1$$

Step (2)

Level of Significance

$$\alpha = 0.05$$

$$1 - \alpha = 0.95$$

Step (3)

$$T_{cal} = \frac{\bar{D}_{x_1 - x_2} - (\mu_2 - \mu_1)}{S_D / \sqrt{n}}$$

Step (4)

$$= \frac{5 - 0}{5.03 / \sqrt{12}}$$

$$T_{cal} = 3.443$$

$$\cancel{3.443} \rightarrow$$

Step (5)

Critical Region:-

$$T_{tab} = +1.796$$

Step (6)

Decision

we will reject H_0 if

$$T_{cal} > 1.796$$

Step (7)

Conclusion

$$3.443 > 1.796$$

We reject H_0

Q4

Same Q(4) in past paper (2017).

Q6

$$n_2 = 72$$

$$\sum x = 1267.2$$

$$\sum x^2 = 22536$$

x	f	fx	fx^2
17.4	12	208.8	3633.12
17.5	16	280	4900
17.6	19	334.4	5885.44
17.7	23	407.1	7205.67
17.8	10	178	3168.4

$$\sum f = 80$$

$$\sum fx = 1408.3$$

$$\sum fx^2 = 24792.63$$

$$n_1 = 80$$

$$\bar{x}_1 = \frac{\sum fx}{\sum f} = \frac{1408.3}{80}$$

$$\boxed{\bar{x}_1 = 17.60}$$

$$\begin{aligned} \sum (x - \bar{x})^2 &= \sum fx^2 - \frac{(\sum fx)^2}{\sum f} \\ &= 24792.63 - \frac{(1408.3)^2}{80} \end{aligned}$$

$$\sum (x - \bar{x})^2 = 1.2688$$

$$s_1^2 = \frac{\sum (x - \bar{x})^2}{\sum f - 1} = \frac{1.2688}{79} = 0.016$$

(2018) -

(20)

$$n_2 = 72, \quad \Sigma X = 1267.2, \quad \Sigma X^2 = 22536$$
$$\Sigma (X - \bar{X})^2 = \frac{\Sigma X^2 - (\Sigma X)^2}{n_2} =$$

$$= \frac{22536 - (1267.2)^2}{72}$$

$$\Sigma (X - \bar{X})^2 = 233.28$$

$$s_2^2 = \frac{\Sigma (X - \bar{X})^2}{n_2 - 1} = \frac{233.28}{71}$$

$$s_2^2 = 3.2856$$

Now,

Combine the two samples to give
give estimate μ and σ^2 .

$$\bar{X}_c = \frac{n_1 \bar{X}_1 + n_2 \bar{X}_2}{n_1 + n_2}$$
$$= \frac{(80)(17.60) + (72)(17.6)}{80 + 72}$$

$$\bar{X}_2 = \frac{\Sigma X}{n_2} = \frac{1267.2}{72}$$

$$\bar{X}_2 = 17.6$$

$$\boxed{\bar{X}_c = 17.6}$$

Now pooled Variance.

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(80 - 1)(0.016) + (72 - 1)(3.2856)}{80 + 72 - 2}$$

$$\boxed{s_p^2 = 1.5636}$$

$$s_p = 1.25$$

(2018)

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Now

90% C-I for μ .

$$1 - \alpha = 90\%$$

$$1 - \alpha = 0.9$$

$$\alpha = 0.1$$

$$\frac{\alpha}{2} = 0.05$$

$$1 - \frac{\alpha}{2} = 1 - 0.05$$

$$1 - \frac{\alpha}{2} = 0.95$$

$$Z_{Tab} = 1.6449$$

$$\bar{x} \pm Z_{Tab} \frac{S_p}{\sqrt{n_1 + n_2}}$$

$$17.6 \pm (1.6449) \frac{1.25}{\sqrt{80 + 72}}$$

$$17.6 \pm 0.1667$$

$$17.433 < \mu < 17.7667$$

$n_1 = 80$
 $n_2 = 72$
 $n_1 + n_2 = 152$

Q7

$$n_1 = 40$$

$$n_2 = 50$$

$$\bar{x}_1 = 70.4$$

$$\bar{x}_2 = 65.3$$

$$S_1^2 = 31.40$$

$$S_2^2 = 44.82$$

Step (1)

$$H_0: \mu_1 - \mu_2 = 2$$

$$H_1: \mu_1 - \mu_2 \neq 2$$

Step (2)

Level of significance :-

$$\alpha = 0.05$$

Step (3)

Test statistical

$$Z_{cal} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Step (4)

Critical region :-

$$\alpha = 0.05$$

$$\frac{\alpha}{2} = 0.025$$

$$1 - \frac{\alpha}{2} = 0.975$$

$$Z_{0.975} = \pm 1.96$$

Step (5)

We will reject H_0 if

$$Z_{cal} > 1.96 \quad \text{or} \quad Z_{cal} < -1.96$$

— (2018) —

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step (b)

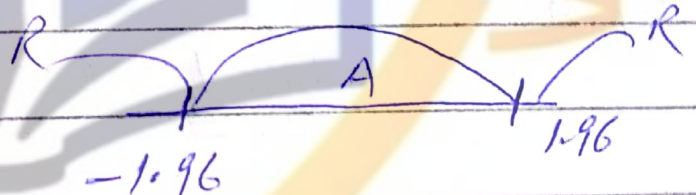
Calculation

$$Z_{\text{cal}} = \frac{(70.4 - 65.3) - 2}{\sqrt{\frac{31.40}{60} + \frac{44.82}{50}}}$$

$$Z_{\text{cal}} = 2.39$$

$$2.39 > 1.96 \quad \text{or} \quad 2.39 < -1.96$$

We ~~cannot~~ reject H_0



PUACP

→ (2018) —

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Q8

<u>Weight Before</u>	<u>Weight after</u>	<u>$D = x_2 - x_1$</u>	<u>D^2</u>
125	136	11	121
195	201	6	36
160	158	-2	4
171	184	13	169
140	145	5	25
201	195	-6	36
170	175	5	25
176	190	14	196
195	190	-5	25
139	145	6	36
		<u>46</u>	<u>673</u>

$$\bar{D} = \frac{\sum D}{n} = \frac{46}{10}$$

$$\boxed{\bar{D} = 4.6}$$

$$\sum (D - \bar{D})^2 = \sum D^2 - \frac{(\sum D)^2}{n}$$

$$= 673 - \frac{(46)^2}{10}$$

$$\sum (D - \bar{D})^2 = 461.4$$

$$\sqrt{\frac{\sum (D - \bar{D})^2}{n-1}} = \sqrt{\frac{461.4}{9}}$$

$$\boxed{S_D = 7.1}$$

Step 1

$$H_0: \mu_D = 0$$

$$H_1: \mu_D \neq 0$$

Step 2 $\alpha = 0.05$

Step 3Test statistical

$$t_{cal} = \frac{\bar{D} - \mu_D}{s_D / \sqrt{n}}$$

Step 4Critical region

$$\frac{\alpha}{2} = 0.025$$

$$\frac{1-\alpha}{2} = 0.975$$

$$v = n - 1$$

$$v = 9$$

$$t_{tab} = 2.262$$

Step 5 DecisionWe will reject H_0 if

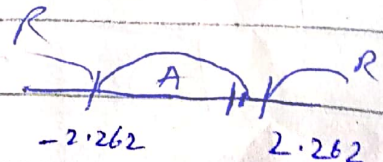
$$t_{cal} > t_{tab}$$

$$t_{cal} < -t_{tab}$$

$$t_{cal} = \frac{4.7 - 0}{7.1 / \sqrt{10}} = 2.09$$

$$2.09 > 2.262$$

$$2.09 < -2.262$$

We do not reject H_0 

(2019)

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Past Paper (2019) -

Q3

(Random Digit)

Height	f	C-f	Str No	$\bar{x}$
60-62	5	5	00-04	61
63-65	18	23	05-22	64
66-68	42	65	23-64	67
69-71	27	92	65-91	70
72-74	8	100	92-99	73
$\Sigma f = 100$				

Using Random Digits

Sr #

Random Samples

Sample Value

$\bar{x}$

(40, 39, 19)	(67, 67, 64)	66
(11, 18, 08)	(64, 64, 64)	64
(86, 39, 80)	(70, 67, 70)	69
(63, 07, 90)	(67, 64, 70)	67
(53, 85, 36)	(67, 70, 67)	68
(98, 37, 21)	(73, 67, 64)	68
(26, 47, 40)	(67, 67, 67)	67
(75, 03, 48)	(70, 61, 67)	66
(75, 32, 32)	(67, 67, 67)	68
(36, 32, 32)	(67, 70, 67)	67

اسی طرح 30 Samples بنائیے۔

— (2019) —

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Q4

$$n=40 \quad \Sigma x = 800 \quad \Sigma x^2 = 16052$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{800}{40}$$

$$\bar{y} = 20$$

$$\Sigma (x - \bar{x})^2 = \Sigma x^2 - \frac{(\Sigma x)^2}{n}$$

$$= 16052 - \frac{(800)^2}{40}$$

$$\Sigma (x - \bar{x})^2 = 52$$

$$s_1^2 = \frac{\Sigma (x - \bar{x})^2}{n-1}$$

$$= \frac{52}{39}$$

$$s_1 = 1.33$$

2nd

<u>x</u>	<u>f</u>	<u>fx</u>	<u>fx²</u>
18	12	216	3888
19	23	437	8303
20	10	200	4000
21	7	147	3087
22	5	110	2420
23	3	69	1587
<u>$\Sigma f = 60$</u>		<u>$\Sigma fx = 1179$</u>	<u>$\Sigma fx^2 = 23285$</u>

$$\Sigma f = 60$$

$$\Sigma fx = 1179$$

$$\Sigma fx^2 = 23285$$

$$\bar{x}_2 = \frac{\Sigma fx}{\Sigma f} = \frac{1179}{60}$$

$$\boxed{\bar{x}_2 = 19.65}$$

$$\begin{aligned} \Sigma (x - \bar{x})^2 &= \Sigma fx^2 - \frac{(\Sigma fx)^2}{\Sigma f} \\ &= 23285 - \frac{(1179)^2}{60} \end{aligned}$$

$$\Sigma (x - \bar{x})^2 = 117.65$$

$$\begin{aligned} s_2^2 &= \frac{\Sigma (x - \bar{x})^2}{\Sigma f - 1} \\ &= \frac{117.65}{59} \end{aligned}$$

$$\boxed{s_2 = 1.99}$$

Now

Combined the two data sets.

$$\bar{x}_c = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

$$= \frac{(40)(20) + (60)(19.65)}{40 + 60}$$

$$\boxed{\bar{x}_c = 19.79}$$

Now,

— (2019) —

Pooled variance is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(39)(1.33) + (59)(1.99)}{40 + 60 - 2}$$

$$s_p^2 = 1.7263$$

$$s_p = 1.3143$$

Now

C-I for μ and σ^2

PUACP

Q6

Sample ~~Q7~~ Q7 in past paper (2018)

$$n = \textcircled{2}_{\text{rows}} \textcircled{5}$$

↓
 $\textcircled{2}$

$\bar{x} = 11$

χ^2 -Test

H_0 : There is No association b/w attributes (Attributes are independent.)

H_1 : There is association b/w attributes.

$\alpha = 0.05$ $1 - \alpha = 0.95$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$\begin{aligned} df &= (r-1)(c-1) \\ &= (2-1)(2-1) \\ &= 1 \end{aligned}$$

$$\chi^2_{\text{Tab}} = \boxed{}$$

we will reject H_0

$$if \chi^2 > \boxed{}$$

f_o = observed freq.

(29)

	<u>A1</u>	<u>W</u>	<u>TOTAL</u>
<u>P.V</u>	80	120	200
<u>Donor went in</u>	170	130	300
	250	250	500

f_e = expected freq

$\frac{200 \times 250}{500} = 100$	$100 = \frac{200 \times 250}{500}$	200
$\frac{300 \times 250}{500} = 150$	$150 = \frac{300 \times 250}{500}$	300
250	250	500

<u>f_o</u>	<u>f_e</u>	<u>$f_o - f_e$</u>	<u>$(f_o - f_e)^2$</u>	<u>$\frac{(f_o - f_e)^2}{f_e}$</u>
80	100	-20	400	4
120	100	20	400	4
170	150	20	400	2.66
130	150	-20	400	2.66

$$\chi^2 = 13.2$$